



Individualized survival prediction and risk stratification using machine learning for patients with malignant struma ovarii: a population-based study with external validation

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Background: Malignant struma ovarii (MSO) is a rare thyroid-type cancer originating in ovarian teratoma. Prognosis of MSO is less studied without unanimous staging or stratification system. This study aimed to develop and validate a machine learning (ML)-based model to predict overall survival (OS) for patients with MSO and to risk-stratify them.

Methods: Patients with histologically confirmed MSO diagnosed in 1975–2021 from the Surveillance, Epidemiology, and End Results (SEER) program were identified as the training cohort. Patients in a systematic literature review were collected as the testing cohort. OS was selected as the outcome, while demographic, clinicopathological and therapeutic information were used as features. Following data encoding, imputing and scaling, univariate feature selection was performed. Cox proportional hazard (CoxPH), Cox with elastic net penalty (CoxNet), random survival forest (RSF), gradient boosting machine (GBM), and survival tree (ST) models were trained and tuned. Each model was evaluated on its c-index, time-dependent area under the curve (AUC), time-dependent Brier score (BS) and stratification ability in the training and the testing cohort respectively. The algorithm that performed the best in the testing cohort was finally chosen for SHapley Additive exPlanations (SHAP) interpretation and Streamlit web application deployment.

Results: The study included 120 and 194 patients in the training and testing cohort respectively. At the end of follow-up (median time 115.5 and 32.5 months respectively), 101 (84.2%) and 181 patients (93.3%) survived respectively. RSF had the best performance in the testing cohort, possessing the highest c-index (0.841, 95% confidence interval: 0.732–0.916), the highest mean AUC (0.852), the lowest integrated BS (0.042), and the smallest P value (<0.001) on log-rank test comparing the stratified groups. According to SHAP, older age, hysterectomy, larger tumor size and more advanced American Joint Committee on Cancer stage had the strongest predictive power for worse OS among all 13 features. An interactive application

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(<https://mso-surv.streamlit.app/>) was then implemented which can display the predicted Kaplan-Meier curve, survival probability, risk stratification and the contributions of features for the output.

Conclusions: We reported the first externally tested time-to-event prognostic prediction model for MSO. ML algorithms enabled precise individual-patient prediction and stratification, and can potentially assist patient counselling and decision-making for treatment and surveillance.

Keywords: Malignant struma ovarii (MSO); machine learning (ML); prediction model; survival analysis; risk stratification

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Introduction

Struma ovarii (SO) is an ovarian teratoma containing >50% thyroid tissue or thyroid-associated malignancy (1). Most SOs are benign, but they can contain histologically malignant components, namely malignant struma ovarii (MSO) (2). MSO is extremely rare with less than 300 cases

reported so far. In a recent hospital-based report, MSO only represents 0.10% of all ovarian teratomas (3). As MSO has unspecific clinical manifestations or imaging appearances, almost all patients are diagnosed and assessed postoperatively (4-6). The complexity of the disease requires multidisciplinary team (MDT) discussion including gynecologists, surgeons, endocrinologists, radiologists and pathologists.

Prognosis of MSO remains controversial with most patients having favorable outcomes but some having lethal disease (7-9). However, given the data paucity there is no widely accepted staging system, hindering its personalized management (10). Using cases extracted from literature, several studies have found age and histology as prognostic factors for MSO (11-15). Nevertheless, due to the limited number of deaths, it is inadequate to draw convincing conclusions. As for MSO confined to the ovary, Ao *et al.* identified ovarian cystectomy as the independent risk factor for disease-free survival (DFS) (15) but another study found no potential risk factors (12).

A few recent studies also attempted to develop risk stratification systems for MSO (16,17). Egan *et al.* adapted 2015 American Thyroid Association (ATA) risk guidelines and proposed a schema for MSO (16). Although high-risk groups were significantly associated with worse overall survival (OS) in National Cancer Database (NCDB) patients, the schema failed to stratify patients in Surveillance, Epidemiology, and End Results (SEER) cohort. With machine learning (ML) algorithms, Alshwayyat *et al.* predicted 5-year survival of MSO patients and achieved a highest accuracy of 80.3% (17), but their study only predicted binary outcomes and was not tested externally. Development of a validated prediction model and risk stratification system may help individualize management and form standard care.

Highlight box

Key findings

- We developed and validated a time-to-event machine learning (ML)-based prognostic tool for patients with malignant struma ovarii (MSO). This model can predict the patients' overall survival (OS) probability at a particular time point and stratify them into high- and low-risk groups. An interactive application was also implemented. Furthermore, explainable ML revealed the association between OS and patient features.

What is known and what is new?

- MSO is an extremely rare cancer arising from thyroid tissue of the ovarian teratoma. Prognosis of MSO seems favorable but remains under-researched. Recent studies attempted to develop prediction systems for MSO but were unsuccessful or not tested externally.
- We used five algorithms to build prognostic prediction models based on Surveillance, Epidemiology, and End Results (SEER) program and validated them in a cohort from a systematic review. Random survival forest had the best performance with a c-index of 0.841 (95% confidence interval: 0.732–0.916). Explainable ML revealed four features most strongly predictive of OS: older age, hysterectomy, larger tumor size and more advanced stage. A user-friendly web application was implemented to visualize and interpret the predicted outcomes.

What is the implication, and what should change now?

- The model has the potential to be translated into clinical practice and help patient counselling, therapeutic and surveillance decision-making. Future studies at a finer scale are needed, especially to incorporate histological, molecular, and genetic information.

Herein, the current study aimed to train and externally validate a prognostic prediction model and risk stratification for MSO using time-to-event ML. We present this article in accordance with the TRIPOD reporting checklist (available at <https://gs.amegroups.com/article/view/10.21037/gs-2025-35/rc>).

Methods

Study population and patients

Flowchart of the study is depicted in *Figure 1A*. Our training dataset used population-based data from National Cancer Institute's SEER program (<https://seer.cancer.gov/>) that represents about 48.0% of the US population (18). Female patients with histologically confirmed MSO [International Classification of Diseases for Oncology, 3rd Edition (ICD-O-3) 9090/3] (19) of the ovary diagnosed in 1975–2021 were included from three databases [Incidence—SEER Research Data, 8 Registries, Nov 2023 Sub (1975–2021); 12 Registries, Nov 2023 Sub (1992–2021); and 17 Registries, Nov 2023 Sub (2000–2021)] of the program and extracted using case listing session of the SEER*Stat (version 8.4.4). All patients' records were reported by hospital or clinic. Dataset of 194 patients with MSO from a previous study (13) that systematically reviewed the literature published in 1957–2021 was used for external testing. Sample size calculation was not performed since all available data were used.

This study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. This study was exempt from ethical review by the Institutional Review Board of Peking Union Medical College Hospital (No. I-24ZM0036) and informed consent was not required as the study utilized only publicly available, de-identified data from previously published literature and SEER database.

Study variables

Given that MSO has good disease-specific survival (DSS) and SEER database does not contain information for DFS, OS was selected as the outcome variable (the label), calculated from diagnosis to death due to all causes. All related fields in SEER database were screened to choose predictor variables (features), including age, tumor-node-metastasis (TNM) categories, American Joint Committee on Cancer (AJCC) stage, extent of disease, grade, tumor

size, chemotherapy, surgical extent, hysterectomy, chemotherapy, and radiotherapy. TNM categories and AJCC stage was defined according to the AJCC version 8 staging system for ovarian cancers (20). Extent of disease was categorized as confined to ovary (CTO) and distant metastasis or peritoneal extension (DM/PE). Tumor grade was categorized as differentiated and poorly-differentiated or undifferentiated (PD/UD). Surgical extent included unilateral salpingo-oophorectomy (USO), bilateral salpingo-oophorectomy (BSO), salpingo-oophorectomy with omentectomy (SOwO) and partial resection (PR; ovarian cystectomy). Hysterectomy and chemotherapy were dichotomous. Radiotherapy included radioiodine (RAI) and external beam radiation therapy (EBRT). For descriptive analysis, we also collected DSS and cause of death.

Data preprocessing and feature selection

Before model training, several steps of data preparation were performed (*Figure 1A*). First, rank variables were ordinally encoded while nominal variables were one-hot encoded. Second, all features with missing values were kept and imputed using an iterative imputer (21). Third, the dataset was scaled by a power transformer (22). The external dataset was then transformed by the fitted imputer and scaler. Lastly, univariate feature selection in the training set was performed. For each algorithm, Harrell's concordance (c-index) of each feature was calculated in 100-times 2-fold cross-validation with 0.5 as the threshold to eliminate the unimportant features.

Model training and evaluation

Cox proportional hazard (CoxPH) (23) and four survival ML algorithms [Cox with elastic net penalty (CoxNet) (24), random survival forest (RSF) (25), gradient boosting machine (GBM) (26) and survival tree (ST) (27)] were used for model training. Optimal hyperparameters for each ML model were chosen by grid search with 5-fold cross-validation, using c-index as the evaluation metric.

Each model's performance was evaluated on the training and external testing dataset. The model performed the best on the testing dataset was determined as the final model. C-index and time-dependent area under the curve (TdAUC) were used to assess models' discrimination. Time-dependent Brier score (TdBS) was used to assess models' calibration. Mean c-index with 95% confidence interval (CI) was calculated by 1,000-times bootstrapping. For TdAUC

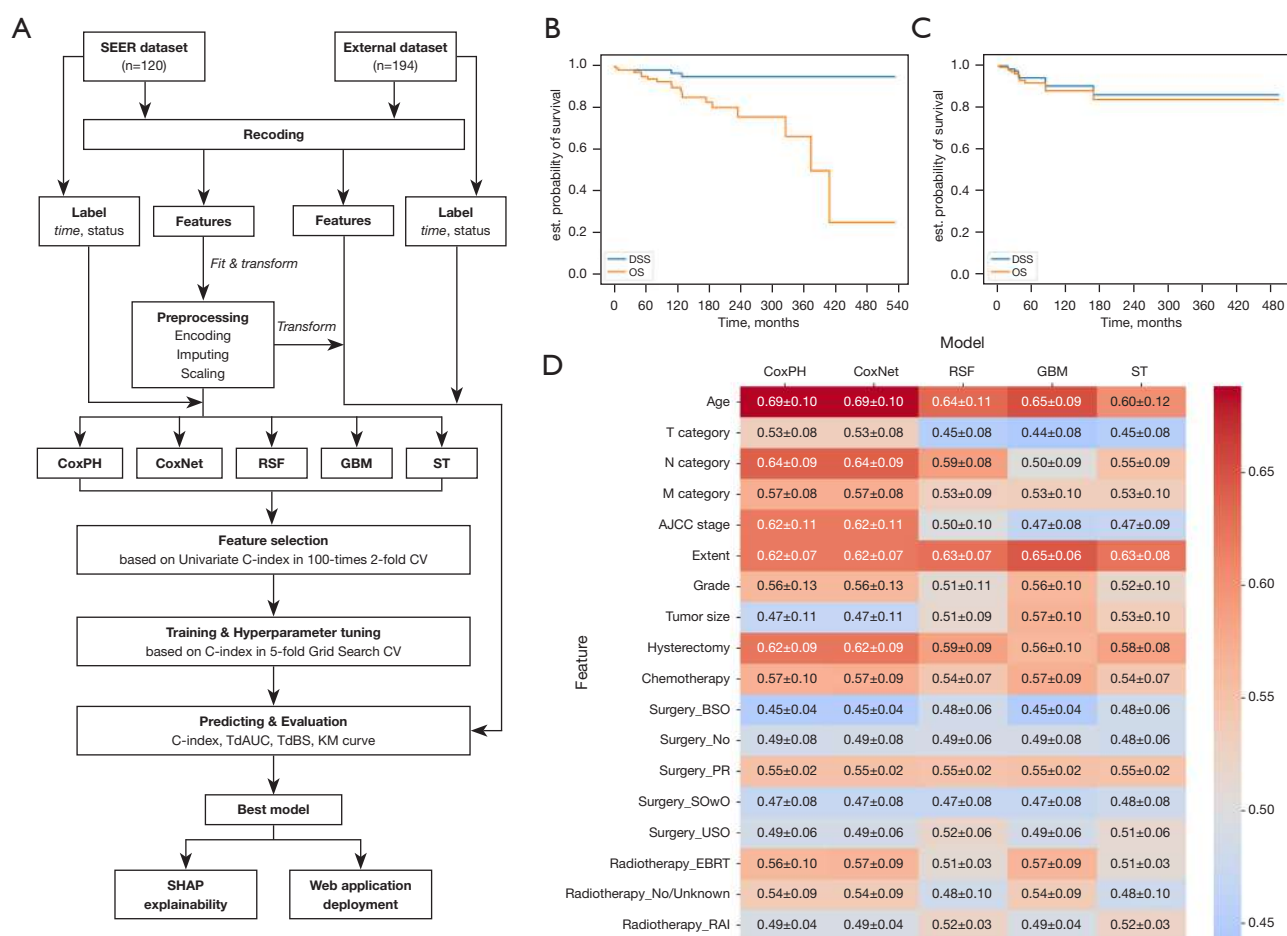


Figure 1 Study overview. (A) Flowchart of the study methodology. (B) KM curves of OS and DSS for patients in the SEER cohort. (C) KM curves of OS and DSS for patients in the external cohort. (D) Heatmap of features' Harrell's concordances (c-indices) in each machine learning model during feature selection. Each c-index was calculated in 100-times 2-fold CV and presented as mean $\pm$ standard deviation. AJCC, American Joint Committee on Cancer; BSO, bilateral salpingo-oophorectomy; CoxNet, Cox with elastic net penalty; CoxPH, Cox proportional hazard; CV, cross validation; DSS, disease-specific survival; EBRT, external beam radiation therapy; est., estimated; GBM, gradient boosting machine; KM, Kaplan-Meier; OS, overall survival; PR, partial resection; RAI, radioiodine; RSF, random survival forest; SEER, Surveillance, Epidemiology, and End Results; SHAP, SHapley Additive exPlanations; SOWO, salpingo-oophorectomy with omentectomy; ST, survival tree; TdAUC, time-dependent area under the curve; TdBS, time-dependent Brier score; TNM, tumor-node-metastasis; USO, unilateral salpingo-oophorectomy.

and TdBS, time points were selected within the 5th–95th percentile ranges of the follow-up time of both cohorts. Mean AUC (mAUC) and integrated Brier score (IBS) were also calculated.

For risk stratification, a risk score was generated for each patient of the cohorts by scikit-survival's "predict" method. Based on the predicted risk scores, a minimal P approach was used to identify the best cut-off value between the 25th and 75th percentile to divide the patients into high- and low-risk groups. P values between the low- and high-risk groups

were calculated by log-rank tests.

Model interpretation and deployment

SHapley Additive exPlanations (SHAP)'s permutation explainer method was utilized to interpret the best model (28). SHAP beeswarm plot was generated to show each feature's contribution on the survival prediction (SHAP value). Features are ranked vertically according to their mean absolute SHAP value. Each dot on a feature line

represents a single patient. Color indicates the feature value, while the position denotes the correlation between feature's contribution and mortality risk.

We also implemented the best model into a Streamlit web application. This application could collect input data interactively and output a SHAP waterfall plot together with the prediction result which includes risk stratification, 5- and 10-year survival probability, and a survival curve. SHAP waterfall plot can display how the input features contribute to the particular prediction.

Statistical analysis

Survival curves and probabilities were estimated by Kaplan-Meier (KM) method. Categorical variables were reported as n (%) and compared by Chi-squared tests. Normally distributed continuous variables were reported as mean $\pm$ standard deviation (SD) and compared with Student *t*-tests, while non-normally distributed ones were reported as median [interquartile range (IQR)] and compared with Mann-Whitney *U* tests. A two-sided *P* value <0.05 was considered statistically significant.

Data recoding and characteristics comparison were performed in R (version 4.4.1), while all other analyses were conducted in Python (version 3.11.5). Model training and evaluation were performed with scikit-learn (29) (version 1.3.2) and scikit-survival (30) (version 0.22.2). Model explanation and deployment was performed with SHAP (28) (version 0.43.0), Streamlit (version 1.28.2) and Streamlit-SHAP (version 1.0.2).

Results

Patient characteristics and survival analysis

One hundred twenty patients from SEER program were included as the training dataset, while 194 patients were included as external testing dataset. Their characteristics are shown in *Table 1*. Median age was 46.5 (IQR, 36.0–59.3) and 46.0 (IQR, 36.0–56.0) years in the SEER and external datasets respectively ($P=0.76$). There was a comparable distribution of tumor staging and extent between two groups with majority of the patients having T1N0M0, stage I tumor confined to the ovary. However, patients in the external cohort had larger tumor sizes (mean 86.5 *vs.* 57.4 mm, $P<0.001$) and more differentiated tumors (95.2% *vs.* 62.5%, $P<0.001$). No significant difference existed in treatment modalities except for radiotherapy, where more

patients received RAI in the external cohort (35.6% *vs.* 10.8%, $P<0.001$).

At the end of follow-up, 101 patients (84.2%) of the training dataset and 181 patients (93.3%) of the external dataset survived ($P=0.02$, *Table 1*). For the SEER cohort, OS was 94.3% at 5 years and 89.1% at 10 years (*Figure 1B*). For the external cohort, OS was 91.4% at 5 years and 87.6% at 10 years (*Figure 1C*). However, OS of the SEER patients continued to decrease (*Figure 1B*), likely because they had a significantly longer median follow-up time than the external cohort (115.5 *vs.* 32.5 months, $P<0.001$).

MSO was the main cause of death in both cohorts, followed by cardiopulmonary diseases (*Table 1*). DSS declined slightly and then flattened in both groups (SEER: 97.5% and 96.0% at 5 and 10 years respectively, *Figure 1B*; external: 93.7% and 89.8% at 5 and 10 years respectively, *Figure 1C*). Given the good DSS, subsequent modelling study was focused on OS.

Model development and evaluation

In univariate feature selection (*Figure 1D*), age was the most important feature in all algorithms except for ST, with the highest c-index in CoxPH and CoxNet (0.69 ± 0.10). Among oncological characteristics, extent of disease exhibited the strongest and most robust predictive value (c-index ≥ 0.62 in all algorithms). Hysterectomy performed better than other treatment characteristics except in GBM. Features with c-index under 0.5 were considered having no predictive value and eliminated before subsequent training of each algorithm. The one-hot encoded features for BSO, no surgery and SOwO were eliminated in all models.

After model training and hyperparameter training, each model was evaluated on the training and testing set respectively. In the training cohort, GBM had the highest c-index (0.966, 95% CI: 0.938–0.988, *Figure S1A*) followed by RSF (0.924, 95% CI: 0.874–0.965). Meanwhile, ST and CoxNet only had slightly higher c-index than CoxPH (0.820, 95% CI: 0.698–0.925). GBM also had the highest mAUC of 0.985 with an almost horizontal TdAUC curve (*Figure S1B*). On the contrary, TdAUC curves of other models declined first and increased from 170 months. RSF performed best in calibration with the lowest IBS of 0.057 (*Figure S1C*). In GBM, CoxNet and CoxPH, TdBS curves increased continuously with time. For risk stratification, each model discriminated patients into low-risk and high-risk group satisfactorily with *P* values <0.001 (*Figure S1D–S1H*). There were negligible little or no overlapping of KM

Table 1 Characteristics of patients with MSO in SEER and external datasets

Variables	SEER (n=120)	External (n=194)	P
Age (years)	46.5 [36.0, 59.3]	46.0 [36.0, 56.0]	0.76
T category ^a			0.12
T1	92 (86.8)	142 (81.1)	
T2	7 (6.6)	8 (4.6)	
T3	7 (6.6)	25 (14.3)	
N category ^b			0.85
N0	94 (97.9)	192 (99.0)	
N1	2 (2.1)	2 (1.0)	
M category ^c			0.06
M0	105 (92.1)	163 (84.0)	
M1	9 (7.9)	31 (16.0)	
AJCC stage ^d			0.12
I	80 (82.5)	142 (73.2)	
II	5 (5.2)	6 (3.1)	
III	3 (3.1)	15 (7.7)	
IV	9 (9.3)	31 (16.0)	
Extent ^e			0.59
CTO	78 (69.6)	142 (73.2)	
DM/PE	34 (30.4)	52 (26.8)	
Grade ^f			<0.001
Differentiated	15 (62.5)	180 (95.2)	
PD/UD	9 (37.5)	9 (4.8)	
Tumor size (mm) ^g	57.4±57.1	86.5±51.8	<0.001
Surgery ^h , n (%)			0.72
USO	24 (22.2)	48 (27.7)	
BSO	42 (38.9)	54 (31.2)	
SOwO	26 (24.1)	46 (26.6)	
PR	9 (8.3)	14 (8.1)	
No	7 (6.5)	11 (6.4)	
Hysterectomy ⁱ			0.20
Yes	54 (52.9)	77 (44.3)	
No	48 (47.1)	97 (55.7)	
Chemotherapy			0.24
Yes	6 (5.0)	18 (9.3)	
No/unknown	114 (95.0)	176 (90.7)	

Table 1 (continued)

Table 1 (continued)

Variables	SEER (n=120)	External (n=194)	P
Radiotherapy			<0.001
RAI	13 (10.8)	69 (35.6)	
EBRT ⁱ	4 (3.3)	5 (2.6)	
No/unknown	103 (85.8)	120 (61.9)	
Follow-up (months)	115.5 [50.3, 196.0]	32.5 [12.0, 74.3]	<0.001
Vital status			0.02
Alive	101 (84.2)	181 (93.3)	
Dead	19 (15.8)	13 (6.7)	
Cause of death			–
MSO	4	10	
Ischemic heart disease	3	–	
Myocardial infarction	–	1	
Diseases of pulmonary circulation	1	–	
COPD	–	1	
Diseases of kidney	1	–	
Multiple myeloma	–	1	
Diseases of head or CNS	1	–	
Thyroid	2	–	
Accidents and adverse effects	3	–	
Other	4	–	
OS (%)			–
5-year	94.3	91.4	
10-year	89.1	87.6	
DSS (%)			–
5-year	97.5	93.7	
10-year	96.0	89.8	

Data are presented as median [IQR], n (%), mean $\pm$ SD, n or percentage. ^a, based on AJCC staging for ovarian tumors. Unknown in 14 (SEER) and 19 (External) patients; ^b, based on AJCC staging for ovarian tumors. Unknown in 24 (SEER) patients; ^c, based on AJCC staging for ovarian tumors. Unknown in six (SEER) patients; ^d, based on AJCC staging for ovarian tumors. Unknown in 23 (SEER) patients; ^e, unknown in eight (SEER) patients; ^f, unknown in 96 (SEER) and five (External) patients; ^g, unknown in 34 (SEER) and 40 (External) patients; ^h, unknown in 12 (SEER) and 21 (External) patients; ⁱ, unknown in 18 (SEER) and 20 (External) patients; ^j, including one patient with both EBRT and RAI. AJCC, American Joint Committee on Cancer; BSO, bilateral salpingo-oophorectomy; CNS, central nervous system; COPD, chronic obstructive pulmonary disease; CTO, confined to ovary; DM/PE, distant metastasis or peritoneal extension; DSS, disease-specific survival; EBRT, external beam radiation therapy; IQR, interquartile range; MSO, malignant struma ovarii; OS, overall survival; PD/UD, poorly differentiated or undifferentiated; PR, partial resection; RAI, radioiodine; SD, standard deviation; SEER, Surveillance, Epidemiology, and End Results; SOWO, salpingo-oophorectomy with omentectomy; TNM, tumor-node-metastasis; USO, unilateral salpingo-oophorectomy.

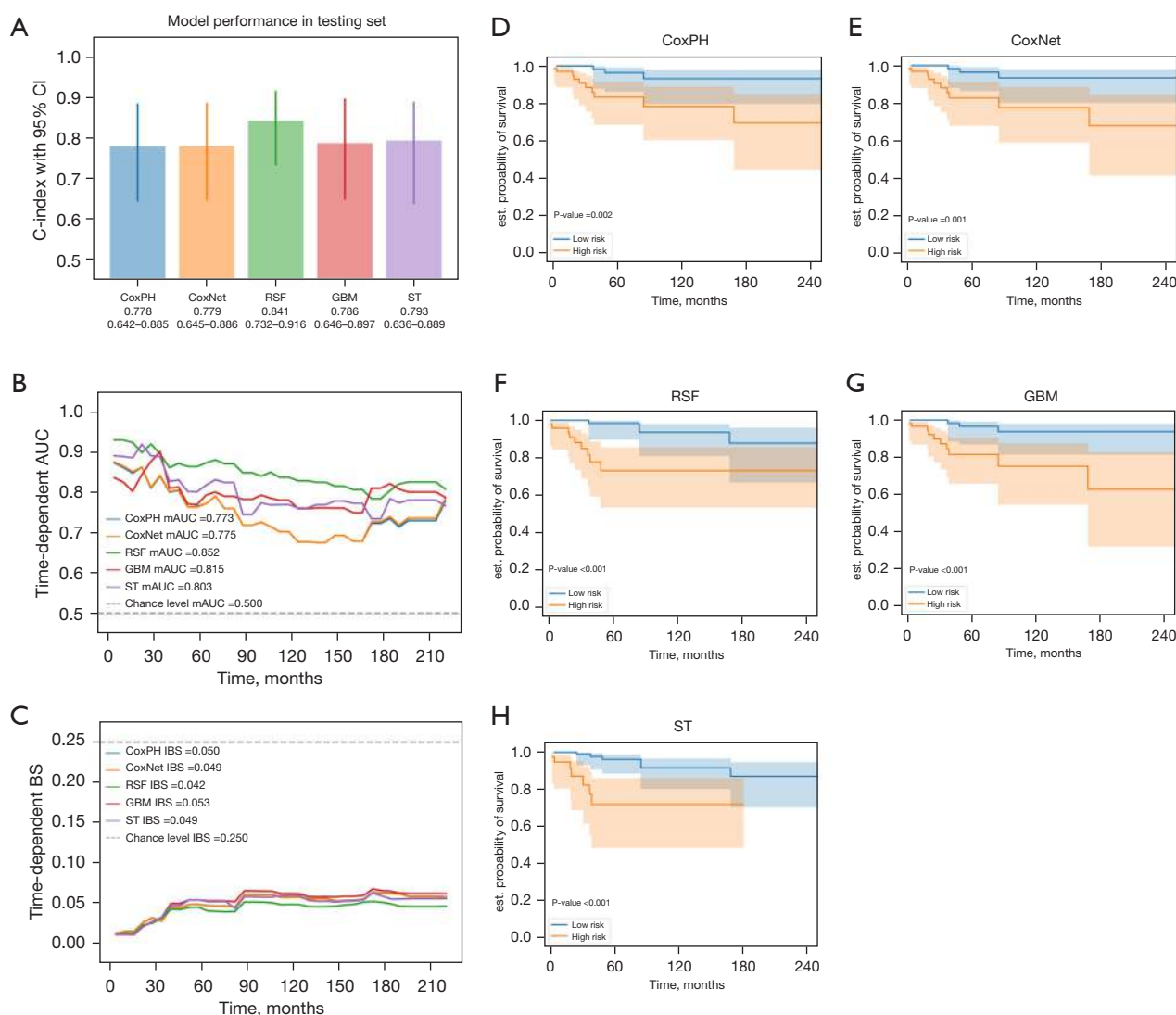


Figure 2 Evaluation of the five models in the external testing set. (A) Bar plot of Harrell's concordances (c-indices) with 95% CIs calculated by 1,000-times bootstrapping. (B) Time-dependent AUC curves in the 5th–95th percentile ranges of the follow-up time. (C) Time-dependent BS curves in the 5th–95th percentile ranges of the follow-up time. (D–H) Kaplan-Meier curves of low-risk and high-risk groups stratified by CoxPH (D), CoxNet (E), RSF (F), GBM (G), and ST (H). P values were calculated by log-rank test. AUC, area under the curve; BS, Brier score; CIs, confidence intervals; CoxNet, Cox with elastic net penalty; CoxPH, Cox proportional hazard; est., estimated; GBM, gradient boosting machine; IBS, integrated Brier score; mAUC, mean area under the curve; RSF, random survival forest; ST, survival tree.

curves' 95% CI.

In the testing cohort, RSF had the most outstanding performance (c-index = 0.841, 95% CI: 0.732–0.916) compared to other models (Figure 2A). Similarly, RSF had the highest mAUC (0.852, Figure 2B) and the lowest IBS (0.042, Figure 2C). TdAUC curves gradually dropped but recovered slightly since 170 months (Figure 2B) while TdBS curves went up with time (Figure 2C). All models

distinguished significantly between low-risk and high-risk groups (Figure 2D–2H) with RSF having the smallest P value (<0.001).

Taken together, these results indicated that RSF should be selected as the final optimal model. As shown by the KM curves for each of the 194 individuals of the testing cohort (Figure 3A), risk stratification based on RSF clearly differentiated patients with higher death risk from those

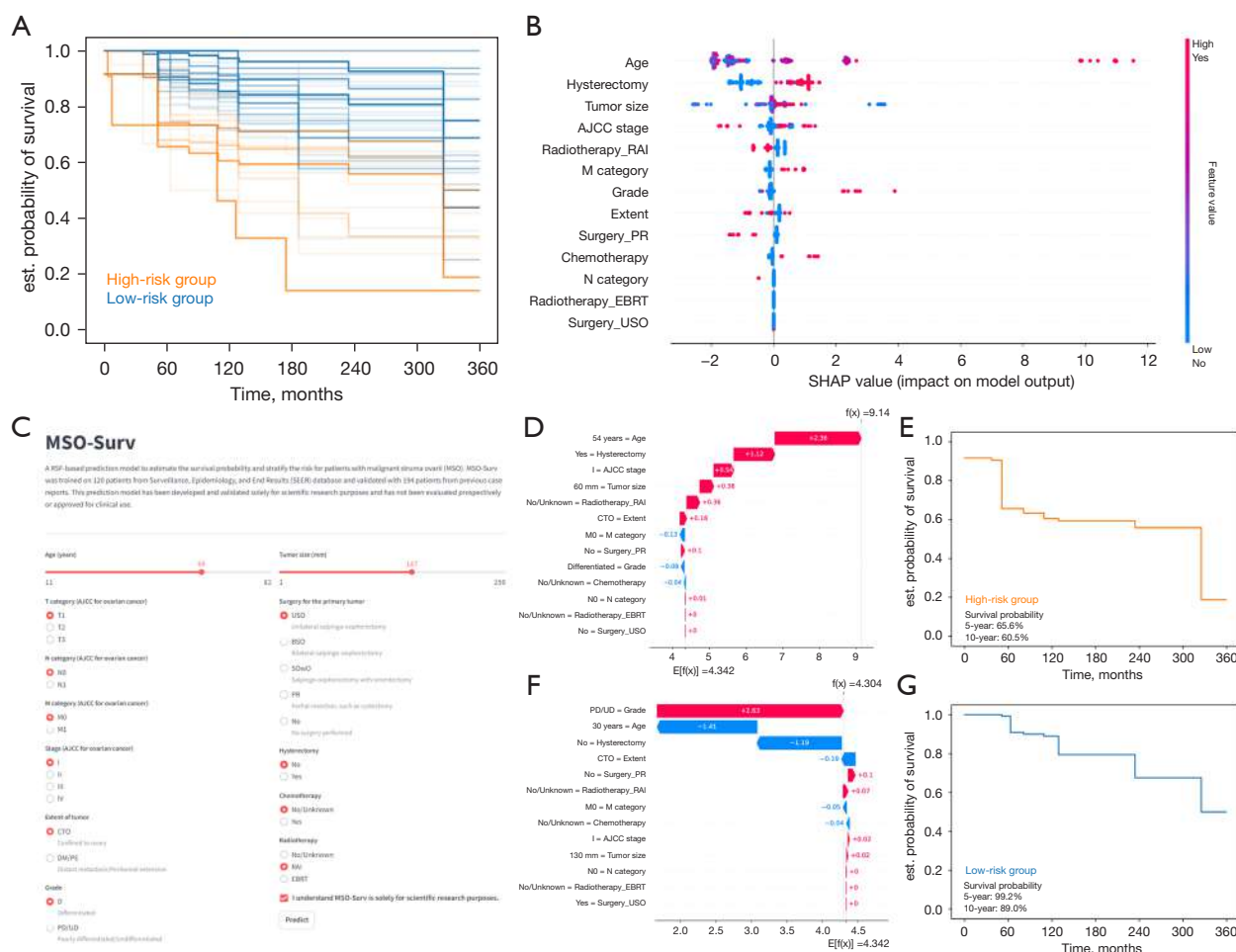


Figure 3 Interpretability and deployment of the final RSF model. (A) KM curves of patients of the testing cohort in the high-risk and low-risk group, stratified by the RSF model. (B) SHAP beeswarm plot of the RSF model. (C) The interface of the web application “MSO-Surv”. (D,E) Illustration of an exemplary output of a 54-year-old patient who was predicted to be at high-risk, with a predicted KM curve plotted (E). (F,G) Illustration of an exemplary output of a 30-year-old patient who was predicted to be at low-risk, with a predicted KM curve plotted (G). The x-axis of the SHAP waterfall plot quantifies each feature’s contribution to MSO-Surv’s output (D,F). AJCC, American Joint Committee on Cancer; BSO, bilateral salpingo-oophorectomy; CTO, confined to ovary; D, differentiated; DM/PE, distant metastasis or peritoneal extension; EBRT, external beam radiation therapy; est., estimated; KM, Kaplan-Meier; MSO, malignant struma ovarii; PD/UD, poorly differentiated or undifferentiated; PR, partial resection; RAI, radioiodine; RSF, random survival forest; SEER, Surveillance, Epidemiology, and End Results; SHAP, SHapley Additive exPlanations; SowO, salpingo-oophorectomy with omentectomy; TNM, tumor-node-metastasis; USO, unilateral salpingo-oophorectomy.

with lower risk.

Model interpretability and web application

The best hyperparameters for the final RSF model were “max_depth” = 680 and “n_estimators” = 4. Unlike traditional CoxPH, there is no coefficient for each predictor variable in RSF, therefore we introduced SHAP visualization to aid in

its interpretability.

Among all 13 features, age had the highest mean absolute SHAP value, indicating its strongest predictive power, followed by hysterectomy, tumor size and AJCC stage (Figure 3B). N category, EBRT and USO had nearly no predictive significance. With the increase in age, tumor size and AJCC stage, the probability for death increased, although there were some outliers. Patients with higher M

category and grade were more likely to be predicted with higher death risk, while the role of disease extent seemed ambiguous. In terms of treatment modalities, receiving RAI and PR seemed associated with lower death probability while receiving hysterectomy and chemotherapy seemed associated with higher probability however there were high missing rate for RAI and chemotherapy.

Lastly, we established a web application for the best model (<https://mso-surv.streamlit.app/>), namely “MSO-Surv”. By inputting the patient’s characteristics in a user-friendly interface (Figure 3C), a SHAP waterfall plot is created implying how the prediction was made (Figure 3D). Meanwhile, MSO-Surv can generate a predicted KM curve with survival probability and risk stratification (Figure 3E). For example, a 54-year-old patient with a 60 mm-sized stage I differentiated tumor who underwent hysterectomy is stratified into the high-risk group with a 5-year and 10-year survival probability of 65.6% and 60.5% respectively (Figure 3D,3E). In contrast, a 30-year-old patient with a 130 mm-sized stage I non-differentiated tumor is stratified into the low-risk group with a 5-year and 10-year survival probability of 99.2% and 89.0% respectively (Figure 3F,3G).

Discussion

Our study constructed and validated a ML-based prognostic tool for patients with MSO. This model can predict the patients’ OS probability at a particular time and stratify them into binary risk groups. An interactive application implementing this model is also available online. Furthermore, explainable ML revealed the association between patient features and the predicted prognosis.

SO is a rare type of ovarian germ cell tumors (OGCTs), while MSO only represents around 0.5–10% of all SOs (31). Furthermore, although MSO is historically malignant, its biological behavior varies considerably, unrelated to its morphologic features (32). In this context, disease course and prognosis of MSO is little understood with evidence mainly from systematic review of case reports/series (11–15,33,34). In 2015, Goffredo *et al.* identified 68 patients with MSO from the SEER program, which was the largest cohort at the time (9). They concluded that both OS and DSS for MSO were excellent whatsoever the treatment administered. The current study provided the updated survival profile of MSO using the latest SEER databases and compared them with a pooled series. We reconfirmed that MSO had an almost horizontal DSS curve (Figure 1B,1C) with age as the most important predictor for OS (Figure 3B),

showing a survival pattern similar to eutopic thyroid cancer (TC).

Recently, statistical and ML-based models have been widely used to predict prognosis of endocrine and gynecological tumors (35–37). In thyroidology, structural recurrence of papillary TC was predicted by ML algorithms which overperformed ATA risk stratification (38). He *et al.* improved the AJCC TNM staging system using decision tree methodology (39). ML models were also applied in OS prediction for anaplastic TC (40). For ovarian tumors, Song *et al.* developed a nomogram to predict 1-, 3-, and 5-year OS of malignant OGCT which included MSO in their study (41). However, very few reports have studied the prediction system exclusively for MSO, probably due to its rarity, which hindered our understanding over the disease as well as personalized managements.

To the best of our knowledge, this is the first study that used time-to-event ML to predict survival of patients with MSO. A recent study by Alshwayyat *et al.* trained a series of ML prognostic models for MSO using data from the SEER program (17). However, their models were only tested internally and could only predict the probability of survival at 5 years. Instead, our study validated the models on an external dataset and yielded robust accuracies. Our models can predict the survival probability for each individual with MSO at all timepoints in a period of more than 25 years. Among all algorithms, RSF reached the highest c-index (0.841, Figure 2A) and mAUC (0.852, Figure 2B), and the lowest IBS (0.042, Figure 2C) in the testing cohort. It is worth noting that RSF also performed well in classification, reaching an AUC of 0.87 for 5-year OS (Figure 2B).

Moreover, there has been no unanimous risk stratification for MSO. Addley *et al.* introduced a risk stratification and management algorithm based on experience of a single institution (6). Another study modified eutopic TC’s risk system for MSO but stratified these patients unsatisfactorily (16). Using a data-driven approach and ML algorithms, our study successfully built a risk system for MSO *de novo* for the first time which can assist decision making and forming guidelines (Figure 3A). Additionally, a user-friendly web application was implemented to enhance its clinical utility (Figure 3C–3G).

We then used SHAP to explain the ML model’s output and to identify MSO’s prognostic risk factors (Figure 3B). Consistent with previous studies, higher tumor grade and distant metastasis was associated with poorer OS. This relationship was also similar for tumor size and AJCC stage, although several outliers were noted which was likely due to

the limited case number. On the other hand, extent (CTO *vs.* DM/PE) had an undetermined ambiguous impact on prediction outcome. Together with a previous study (13), our result questioned whether CTO [AJCC/International Federation of Gynecology and Obstetrics (FIGO) stage I] itself plays a role as a prognostic factor. Also, N category had nearly no predictive power for OS, suggesting that staging or therapeutic lymph node dissection could be omitted in MSO. Future efforts should be made to develop a new staging system for MSO as it has a different survival profile from other ovarian tumors and the traditional AJCC/FIGO system might not be suitable.

The optimal extent of resection for MSO remains under debate (15,31). Unexpectedly, we have found that hysterectomy was associated with poorer OS while PR was associated with better OS. Similar to other malignant ovarian germ cell tumors (MOGCTs) (42,43), a conservative surgery might be considered for MSO in order to preserve fertility and improve quality of life, since our data showed no prognostic benefits from extended resection. However, this observation might likely also reflect selection bias, where hysterectomy may have been performed more frequently in advanced cases. Definitive treatment recommendations should await validation in larger prospective studies. We also found that chemotherapy was negatively associated with OS, implying that MSO might be treated as a different entity from other MOGCTs where platinum-based regimen is a major treatment (44). Contrarily, our results showed RAI was related to improved survival, possibly supporting the positive role of thyroid-related treatment. Though postoperative treatment for MSO was not standardized, total thyroidectomy with subsequent RAI may be recommended for selected patients especially those with distant metastasis or synchronous eutopic TC (45). However, these observations require careful interpretation due to high rate of missingness (Table 1). We emphasize these therapy-related associations should be viewed as hypothesis-generating rather than conclusive and need future validation.

Taken together, although MSO is mainly treated by gynecologists and diagnosed postoperatively, its biological features may resemble an eutopic TC. MDT involving endocrine surgeons and nuclear radiologists should be considered for all patients with MSO. And in this case, our study provided a helpful tool for patient consultation, therapeutic decision making and tailoring follow-up plan. While future large-scale studies are needed for validation, at this stage our prognostic model may serve as a valuable

counseling tool in visualizing outcomes, informing modifiable/non-modifiable factors and managing follow-up expectations.

There are some limitations of this study. Firstly, the sample size was small for a modelling study which may introduce some biases (46). Nevertheless, our study attempted to include all available cases and was one of the largest series so far, given the extremely rare incidence of the tumor. Secondly, similar to other SEER-based studies (47), some features had a high proportion of missing values, such as RAI and chemotherapy. To address this problem, we used multiple imputation and external validation to ensure reliability, but residual confounding may persist, rendering our associations suggestive rather than conclusive. Thirdly, some therapeutic, histological and molecular features of MSO were not included in this study due to SEER database constraints. Unfortunately, we were not able to assess the impact of total thyroidectomy and surgical margin on MSO OS. Recent efforts have found some genetic mutations associated with malignancy and aggressiveness of MSO (48,49), however their prognostic values are still to be studied. Lastly, although SHAP technique was used to explain the ML model, it was not designed for causal inference. Research is warranted to further uncover the underlying mechanisms in MSO tumorigenesis and progression. Meanwhile, results of our study should be verified in a larger population and with longer follow-up. Future multi-center or national hospital-based registry studies may incorporate histopathological, molecular, and genetic profiles to investigate this disease at a finer scale.

Conclusions

In conclusion, we successfully developed the first externally validated time-to-event model to predict prognosis of individuals with MSO. A web-based system named MSO-Surv was deployed integrating probability prediction, risk stratification, and interpretability. We identified the most significant predictive features for worse OS, which include older age, hysterectomy, larger tumor size and higher AJCC stage. The presented model has the potential to be translated into clinical practice, to aid in patient counselling and designating treatment or surveillance plan.

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Footnote

Reporting Checklist: The authors have completed the TRIPOD reporting checklist. Available at <https://gs.amegroups.com/article/view/10.21037/gS-2025-35/rc>

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Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. This study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. This study was exempt from ethical review by the Institutional Review Board of Peking Union Medical College Hospital (No. I-24ZM0036) and informed consent was not required as the study utilized only publicly available, de-identified data from previously published literature and SEER database.

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